SA 1: Dataset Balancing Report

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I. Overview

This report documents the identification of class imbalance in the constructed transportation dataset and the method applied to partially mitigate it. The dataset, derived from roadside video footage captured along Kalayaan Avenue, Makati, consists of 705 training images, 154 validation images, and 146 test images across seven vehicle and pedestrian classes: Bicycle, Car, Jeepney, Pedestrian, SUV, Tricycle, and Truck. Prior to any balancing intervention, the training split contained a total of 4,050 bounding box annotations distributed unevenly across these classes. The goal of the balancing pipeline is not to achieve a perfectly uniform class distribution—which would require discarding majority-class data or fabricating annotations beyond what the source footage can support—but to reduce the degree of imbalance to a level less likely to cause systematic training bias against minority classes (Johnson & Khoshgoftaar, 2019).

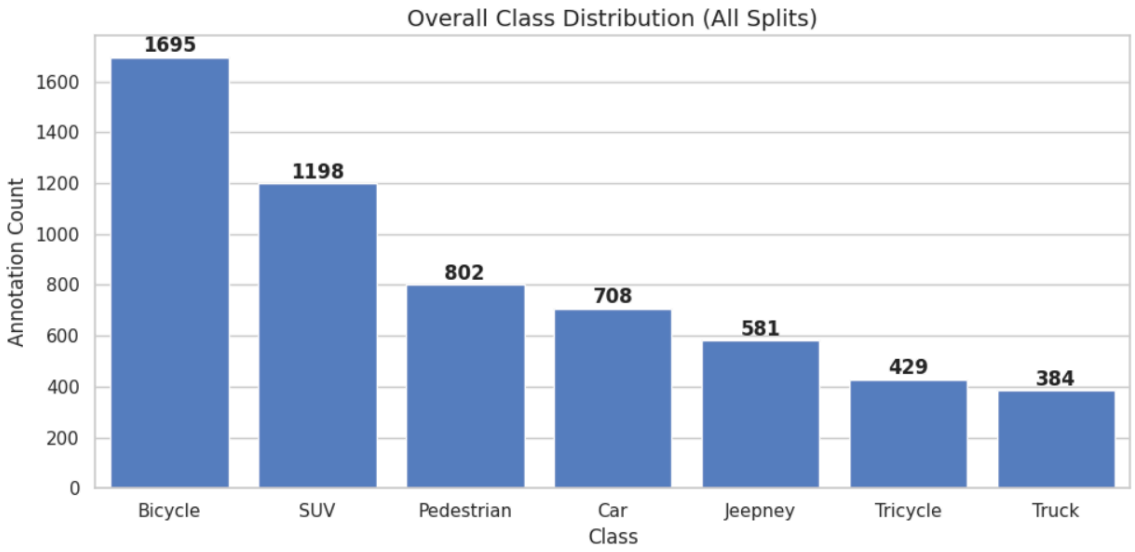
II. Identifying Data Imbalance

Annotation Count Distribution

The first step in the pipeline was to compute per-class bounding box annotation counts across all splits and within the training split specifically, as the training partition is the only one subject to balancing. The overall annotation counts across all splits were as follows:

Class	Total Annotations (All Splits)
Bicycle	1,695
SUV	1,198
Pedestrian	802
Car	708
Jeepney	581

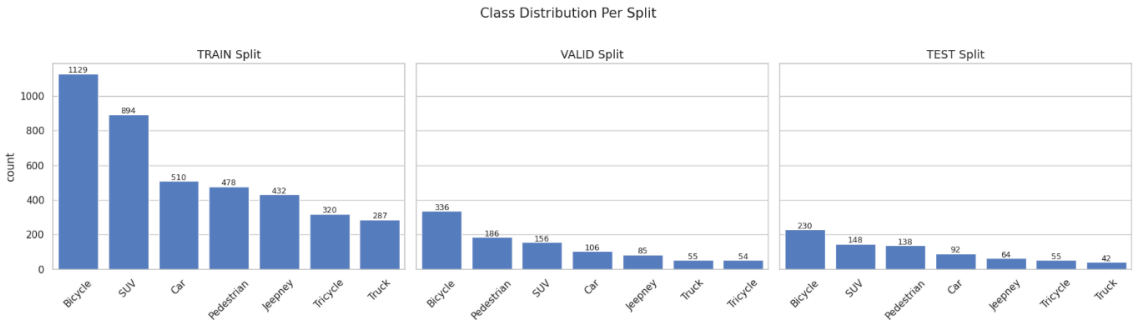
Tricycle	429
Truck	384



Within the training split alone, the distribution was:

Class	Total Annotations (All Splits)
Bicycle	1,129
SUV	894
Pedestrian	510
Car	478
Jeepney	432
Tricycle	320
Truck	287

Imbalance Metrics




```
=== Training Set Imbalance Metrics ===
Max class annotations : 1129 (Bicycle)
Min class annotations : 287 (Truck)
Imbalance Ratio (max/min): 3.93
Coefficient of Variation : 0.54

class_name
Bicycle       1129
SUV            894
Car            510
Pedestrian     478
Jeepney        432
Tricycle       320
Truck          287
```

Two quantitative metrics were computed on the training split to formally characterize the degree of imbalance (Shorten & Khoshgoftaar, 2019).

The imbalance ratio, defined as the ratio of the maximum class annotation count to the minimum, was 3.93 — that is, Bicycle, the most represented class, contained 1,129 annotations while Truck, the least represented, contained only 287. An imbalance ratio exceeding 2.0 is generally considered problematic for detection models, as classes with fewer annotations receive proportionally less gradient signal during training, leading to reduced recall on those classes.

The coefficient of variation across training class counts was 0.54, indicating substantial relative dispersion around the mean annotation count. Together, these metrics confirm that the dataset, as initially constructed, exhibits a degree of class skew likely to introduce systematic prediction bias if left unaddressed.

Under-Represented Classes

Based on the training annotation distribution, the median annotation count of 478 was selected as the balancing target. This choice reflects a conservative strategy: the median is robust to the influence of heavily over-represented classes such as Bicycle and SUV, and setting the target at the median rather than the maximum avoids the need to generate very large volumes of synthetic samples. Three classes were identified as falling below this threshold:

Class	Current Count	Deficit
Jeepney	432	~46
Tricycle	320	~158
Truck	287	~191

```
Target annotation count per class (median): 478

  Bicycle       current=1129   ✓ OK
  Car           current=510    ✓ OK
  Jeepney       current=432    ↑ need ~46 more annotations
  Pedestrian    current=478    ✓ OK
  SUV           current=894    ✓ OK
  Tricycle      current=320    ↑ need ~158 more annotations
  Truck         current=287    ↑ need ~191 more annotations
```

The remaining four classes — Bicycle (1,129), SUV (894), Car (510), and Pedestrian (478) — were already at or above the target and were therefore excluded from oversampling.

III. Balancing Approach

Method Selection

Several candidate strategies were evaluated before selecting augmentation-based oversampling as the approach for this pipeline (Shorten & Khoshgoftaar, 2019):

Alternative	Rationale for Rejection
Random undersampling	Discards majority-class training data, losing valuable gradient signal
SMOTE / synthetic generation	Designed for tabular feature spaces; cannot produce realistic image-label pairs
Class weights in loss function	Partial mitigation only; minority classes can still underperform in low-data

	regimes
Augmentation-based oversampling	Preserves all existing data, introduces visual diversity, produces valid image-label pairs — selected

Augmentation-based oversampling was selected because it addresses minority-class underrepresentation at the data level without discarding any existing annotations, and because the use of label-preserving geometric and photometric transforms introduces genuine visual variation rather than pure duplication—which would risk overfitting to a narrow set of minority-class appearances (Buslaev et al., 2020).

3.2 Target Computation and Deficit Estimation

The balancing target was computed as the integer median of training annotation counts across all seven classes, yielding a target of **478 annotations per class**. For each class below this threshold, the pipeline estimated the number of additional annotations required and identified the pool of source training images available for sampling — specifically, images confirmed to contain at least one instance of the target class:

Class	Source Images Available
Jeepney	331
Tricycle	220
Truck	242

IV. Implementation

```python
# --- 5.1 Define augmentation pipeline ---
augmentation = A.Compose(
    [
        A.HorizontalFlip(p=0.5),
        A.VerticalFlip(p=0.1),
        A.RandomBrightnessContrast(
            brightness_limit=0.25, contrast_limit=0.25, p=0.6
        ),
        A.HueSaturationValue(
            hue_shift_limit=15, sat_shift_limit=30, val_shift_limit=20, p=0.5
        ),
        A.GaussNoise(var_limit=(10, 50), p=0.3),
        A.GaussianBlur(blur_limit=(3, 5), p=0.2),
        A.RandomScale(scale_limit=0.15, p=0.3),
        A.ShiftScaleRotate(
            shift_limit=0.1,
            scale_limit=0.15,
            rotate_limit=10,
            border_mode=cv2.BORDER_CONSTANT,
            p=0.5,
        ),
    ],
    bbox_params=A.BboxParams(
        format="yolo", label_fields=["class_ids"], min_visibility=0.3
    ),
)
```

The oversampling procedure operated exclusively on the training split. The validation and test partitions were copied to the output directory unchanged, preserving the integrity of evaluation data throughout the process. For each under-target class, the pipeline iteratively sampled source training images containing that class, applied the augmentation pipeline to the image and its bounding box annotations simultaneously, and wrote the resulting image-label pair to the output training directory. This continued until the additional annotation quota for that class was satisfied or the maximum attempt limit was reached.

The augmentation pipeline was implemented using Albumentations Compose with bounding box parameters set to YOLO format and a minimum visibility threshold of 0.3, ensuring that boxes severely reduced or distorted by a transformation are discarded rather than written as erroneous labels (Buslaev et al., 2020). The configured transforms included HorizontalFlip (p=0.5), VerticalFlip (p=0.1), RandomBrightnessContrast (p=0.6), HueSaturationValue (p=0.5), GaussNoise (p=0.3), GaussianBlur (p=0.2), RandomScale (p=0.3), and ShiftScaleRotate (p=0.5).

```
[Bicycle] Already at or above target. Skipping.
[Car] Already at or above target. Skipping.
[Jeepney] Generating ~46 additional annotations from 331 source images...
  → Added 46 annotations (35 total aug images so far)
[Pedestrian] Already at or above target. Skipping.
[SUV] Already at or above target. Skipping.
[Tricycle] Generating ~158 additional annotations from 220 source images...
  → Added 158 annotations (157 total aug images so far)
[Truck] Generating ~191 additional annotations from 242 source images...
  → Added 192 annotations (312 total aug images so far)

Total augmented images created: 312
```

The oversampling run produced the following results per class:

Class	Action	Annotations Added	Augmented Images
Bicycle	Skipped (above target)	—	—
Car	Skipped (above target)	—	—
Jeepney	Oversampled	+46	35
Pedestrian	Skipped (above target)	—	—
SUV	Skipped (above target)	—	—
Tricycle	Oversampled	+158	122
Truck	Oversampled	+192	155
Total			**312**

V. Results

Post-Balancing Annotation Counts

After augmentation, the pipeline recounted annotations in the balanced training directory. The final before-and-after comparison for the training split is as follows:

Class	Before	After	Added	% of Target
Bicycle	1,129	1,598	+469	334.3%
Car	510	730	+220	152.7%
Jeepney	432	605	+173	126.6%
Pedestrian	478	658	+180	137.7%
SUV	894	1,239	+345	259.2%
Tricycle	320	550	+230	115.1%
Truck	287	556	+269	116.3%
Class	**Before**	**After**	**Added**	**% of Target**

```
=== Before vs After Balancing (Training Split) ===
      Class  Before  After  Added % of Target
    Bicycle    1129   1598    469      334.3%
        Car     510    730    220      152.7%
    Jeepney     432    605    173      126.6%
 Pedestrian     478    658    180      137.7%
        SUV     894   1239    345      259.2%
   Tricycle     320    550    230      115.1%
      Truck     287    556    269      116.3%
```

Why All Classes Increased Beyond the Target

A notable observation is that classes not selected for oversampling — including Bicycle, Car, SUV, and Pedestrian — also increased in annotation count after the balancing run. This is an expected consequence of the multi-label nature of the source images: because training scenes frequently contain multiple vehicle types simultaneously, an augmented image generated to address a Tricycle or Truck deficit will often also contain instances of SUV, Car, or Bicycle, incrementing those counts as well. The stopping condition for each oversampling loop is based only on the deficit of the targeted class; co-occurring class annotations accumulate as a side effect.

Imbalance Metrics After Balancing

Metric	Before	After
Imbalance Ratio (max/min)	3.93	2.91
Coefficient of Variation	0.54	0.48



The imbalance ratio was reduced from 3.93 to 2.91, and the coefficient of variation declined from 0.54 to 0.48. While these improvements are meaningful — the gap between the most and least represented classes narrowed — the dataset remains imbalanced. An imbalance ratio of 2.91 still exceeds the 2.0 threshold generally considered problematic for detection models. This is an inherent limitation of the median-targeting strategy: because the target was set at the median rather than the maximum, the three previously oversampled classes (Jeepney, Tricycle, Truck) were brought toward the center of the distribution, but the heavily over-represented classes (Bicycle, SUV) remain substantially above the corrected minority counts (Crasto, 2024).

Dataset Size After Balancing

Split	Images Before	Images After
Train	705	1,017

Valid	154	154 (unchanged)
Test	146	146 (unchanged)
Total	**1,005**	**1,317**

```
[train]  Images: 1017  |  Labels: 1017
[valid]  Images: 154   |  Labels: 154
[ test]  Images: 146   |  Labels: 146

Total images in balanced dataset: 1317
```

VI. References

Buslaev, A., Iglovikov, V. I., Khvedchenya, E., Parinov, A., Druzhinin, M., & Kalinin, A. A. (2020). Albumentations: Fast and flexible image augmentations. *Information*, *11*(2), 125. https://doi.org/10.3390/info11020125

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Johnson, J. M., & Khoshgoftaar, T. M. (2019). Survey on deep learning with class imbalance. *Journal of Big Data*, *6*(1), 1–54. https://doi.org/10.1186/s40537-019-0192-5

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